

Software Requirements Specification (SRS)

Pixous Estimation & Invoicing Platform — Version 1.0

Document status: Baseline for shipped V1 functionality

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1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for Version 1 of the Pixous Estimation & Invoicing Platform. It describes the system **as built**, so that it can serve as (a) a reference for QA sign-off, (b) an onboarding reference for new engineers, and (c) a baseline against which V2 scope can be planned.

1.2 Product Overview

The platform is an internal tool for Pixous Technologies that automates two related workflows for a software services business:

1. **Estimation Maker** — ingests a client's requirement document (PDF, DOCX, plain text, or a URL), uses an AI pipeline to analyze scope and generate a cost/timeline estimate, and produces client-ready Quotation, Business Requirements Document (BRD), and Software Requirements Specification (SRS) documents.
2. **Invoice Maker** — generates a branded, GST-compliant tax invoice either from a completed estimation or from manually entered line items, tracks its payment status, and exports it as a PDF.

The system is single-tenant (one company profile), India-specific (GST/INR throughout), and intended for internal staff use only — there is no public-facing or client-facing account access in V1.

1.3 Intended Audience

Engineering team, QA, and Pixous management evaluating the tool for internal rollout.

1.4 Scope of Version 1

V1 covers: authenticated internal users, AI-assisted estimation generation, manual (non-AI) estimation and invoicing, document editing and PDF export, company branding/letterhead, a fixed developer rate card, and a basic analytics dashboard. Any authenticated internal user has equal access to all of the above in V1 — there is no role-based access control (see §2.2 and §7). Multi-currency, client self-service portals, payment gateway integration, and horizontal scaling of the job pipeline are explicitly **out of scope** for V1 (see §7).

2. Overall Description

2.1 System Architecture

- **Backend:** Python / FastAPI, serving both the REST API and the built frontend as a single deployable unit.

- **Frontend:** React 18 + TypeScript (Vite), React Router, Tailwind CSS.
- **Database:** SQLAlchemy ORM; SQLite for local/dev, PostgreSQL in production.
- **AI pipeline:** LangGraph-orchestrated multi-agent workflow using Mistral (OCR), Groq (LLM analysis/estimation/document generation), and Google Gemini (web-grounded market pricing research).
- **Document export:** WeasyPrint / Playwright (headless Chromium) for Markdown/HTML → PDF conversion.

2.2 User Roles

In V1, all internal user accounts have equal, undifferentiated access — any authenticated user can create, edit, and delete estimations and invoices, manage organization/branding, and update the rate card. There is no role-based access control (RBAC) in V1; it is a candidate for future development if the business later needs to separate duties (e.g. restricting financial edits to a Finance function, or organization/branding changes to an Admin function) — see §7.

There is no self-service sign-up; accounts are provisioned by an operator via a CLI script.

2.3 Operating Environment

Single company/tenant deployment. Currency fixed to INR; tax model fixed to Indian GST. Designed to run as one backend process (see §6.3 for the scaling constraint this implies).

3. Functional Requirements

3.1 Authentication & Session Management

- FR-1.1: Users shall log in with a username and password; the system returns a signed, time-limited session token (24-hour expiry).
- FR-1.2: The system shall lock out an IP+username pair for 15 minutes after 5 failed login attempts within a 15-minute window.
- FR-1.3: The system shall reject expired or tampered tokens on every authenticated request.
- FR-1.4: An unauthenticated endpoint shall expose company branding (logo/name) so the login screen can display it pre-authentication.
- FR-1.5: A machine-to-machine API key (`X-API-Key`) mode shall be supported for automated testing tools, resolving to a designated internal identity with the same permissions as a logged-in user.
- FR-1.6: Passwords shall be stored hashed (bcrypt); any legacy plaintext password shall be transparently upgraded to a hash on next successful login.

3.2 Authorization

- FR-2.1: Any authenticated internal user shall be able to perform any action in the system (create/edit/delete estimations and invoices, manage organization/branding, update the rate card) — V1 has no role-based access control.
- FR-2.2: Estimation and invoice mutations shall be recorded in an audit log (actor, action, before/after values, timestamp), regardless of the acting user's account.

3.3 Document Ingestion (Estimation Maker)

- FR-3.1: The system shall accept a requirement source as an uploaded file (PDF, DOCX), a URL, or pasted plain text.

- FR-3.2: PDF sources shall be routed through OCR to extract text and structure; DOCX sources shall be parsed directly.
- FR-3.3: URL ingestion shall reject non-HTTP(S) schemes and any address resolving to a private, loopback, link-local, or otherwise reserved IP range, shall not follow redirects, and shall cap downloaded content at 50MB (SSRF hardening).

3.4 AI-Assisted Analysis & Estimation

- FR-4.1: The system shall extract a structured project analysis from ingested text: client name, project name/description/type, technology stack, a granular list of requirements (each with category, priority, complexity, estimated hours, required role, dependencies), assumptions, risks, and explicit out-of-scope items.
- FR-4.2: The system shall produce a cost and timeline estimate that maps requirement effort to roles and hours, priced against the current developer rate card.
- FR-4.3: The system shall research current market pricing (infrastructure, licensing, third-party services) via web-grounded search and incorporate it into the estimate in INR.
- FR-4.4: The system shall optionally generate a full BRD (business goals, stakeholders, scope) and/or SRS (technical architecture, data schema, API surface, functional flows, non-functional requirements, and an entity-relationship diagram) alongside the Quotation, at the user's discretion per job.
- FR-4.5: The system shall assemble a client-facing Quotation document deterministically from the analysis, estimation, and market-research outputs (no LLM call at this stage, to guarantee consistent totals).

3.5 Job Lifecycle

- FR-5.1: Submitting a source shall start an asynchronous job and return a job identifier immediately.
- FR-5.2: The system shall expose job progress (current pipeline stage, log, and final result) via a polling endpoint.
- FR-5.3: The system shall additionally expose a synchronous long-poll endpoint (bounded timeout) for load-testing tools that require a single blocking call.
- FR-5.4: A running job shall be cancellable by the user who owns the session.
- FR-5.5: On successful completion, the system shall persist the client, estimation, and generated documents (Quotation/BRD/SRS) to the database.

3.6 Estimation Management

- FR-6.1: Users shall be able to list all estimations, grouped by client, with status, totals, and linked invoice metadata.
- FR-6.2: Users shall be able to view the full underlying pipeline data for an estimation.
- FR-6.3: Users shall be able to create a **manual estimation** (client name, project name, and a list of line items with description/quantity/rate) without invoking the AI pipeline, for cases where no requirement document exists.
- FR-6.4: Any authenticated user shall be able to edit an estimation's project name, timeline, and grand total; concurrent edits shall be protected by optimistic locking (a version conflict shall be rejected rather than silently overwritten).
- FR-6.5: Any authenticated user shall be able to delete an estimation; deletion shall be a soft delete (recoverable/auditable), not a physical row removal.

- FR-6.6: Estimation status shall progress through Draft → Processing → Completed/Failed → Approved/Sent → Archived.

3.7 Document Editing & Export

- FR-7.1: Users shall be able to view and edit the Markdown content of a Quotation, BRD, or SRS document; each edit shall be preserved as a new version rather than overwriting history.
- FR-7.2: Users shall be able to download any Quotation/BRD/SRS/Invoice as a PDF, matching the on-screen rendering (including diagrams where the export format supports it).
- FR-7.3: Any invoice HTML content submitted for edit shall be sanitized to remove executable script content before storage.

3.8 Invoice Management (Invoice Maker)

- FR-8.1: Users shall be able to generate an invoice from a completed estimation, specifying tax percentage and payment due period.
- FR-8.2: Users shall be able to create an invoice directly from manually entered line items, independent of any estimation.
- FR-8.3: Each invoice shall compute subtotal, GST amount, discount, and grand total, and shall carry a unique sequential invoice number.
- FR-8.4: Any authenticated user shall be able to update invoice status (Draft, Sent, Paid, Partially Paid, Overdue, Cancelled); marking an invoice Paid shall auto-stamp the payment date.
- FR-8.5: Any authenticated user shall be able to directly patch invoice financial fields (amounts, status, payment date); such changes shall be audit-logged.
- FR-8.6: Any authenticated user shall be able to hard-delete an invoice; if no invoices remain for its parent estimation, the estimation's status shall revert accordingly.
- FR-8.7: Users shall be able to list invoice history and view/download any invoice as branded PDF.

3.9 Organization Profile & Branding

- FR-9.1: The system shall maintain a single company profile: name, tagline, address, contact details, GSTIN, registration number, certifications, bank account details, and authorized signatory.
- FR-9.2: Any authenticated user shall be able to update the company profile, with server-side validation of email, phone, and GSTIN formats.
- FR-9.3: Any authenticated user shall be able to upload/replace/remove branding assets (logo, signature, seal), limited to 5MB, with the server verifying that uploaded image content genuinely matches its declared file type.
- FR-9.4: Any authenticated user shall be able to retroactively re-apply the current branding to all previously generated invoices and documents in one action.
- FR-9.5: Branding assets shall be durably stored (backed up in the database) so they survive a redeploy of the application filesystem.

3.10 Rate Card

- FR-10.1: The system shall maintain a fixed list of developer roles and their hourly billing rates, used by every estimation.
- FR-10.2: Any authenticated user shall be able to add, update, or deactivate a role's rate; rate changes shall be versioned (old rates retained with an end date, not deleted) so historical estimations remain traceable to the rate in effect at the time.

3.11 Analytics

- FR-11.1: The system shall present an aggregate dashboard: total estimations/invoices, counts for the current day/month, revenue paid vs. pending, invoice status breakdown, and recent activity.

3.12 Clients

- FR-12.1: The system shall derive a client list from estimation history and allow browsing estimations grouped by client.
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4. Non-Functional Requirements

4.1 Security

- NFR-1.1: All API endpoints except login, token validation, and public branding shall require authentication.
- NFR-1.2: Session tokens shall be cryptographically signed and shall be rejected if the embedded identity does not correspond to a real, currently valid account.
- NFR-1.3: All destructive or financially significant mutations (estimation edit/delete, invoice edit/delete, organization/rate-card changes) shall be captured in an audit trail, though in V1 they are not further restricted by role — any authenticated user may perform them.
- NFR-1.4: User-supplied HTML (invoice content) shall never be persisted or rendered with executable script content.
- NFR-1.5: Outbound URL ingestion shall be hardened against SSRF (see FR-3.3).
- NFR-1.6: Uploaded images shall be validated against spoofed file extensions.

4.2 Reliability & Data Integrity

- NFR-2.1: Concurrent edits to the same estimation shall not silently overwrite one another (optimistic locking).
- NFR-2.2: Deleted estimations shall be recoverable in principle (soft delete) rather than immediately and irreversibly purged.
- NFR-2.3: Document edits (Quotation/BRD/SRS) shall retain version history.

4.3 Usability

- NFR-3.1: Long-running estimation jobs shall show step-by-step progress so a user can track pipeline stage (ingestion → OCR → analysis → estimation → web research → BRD → SRS → quotation).
- NFR-3.2: A user shall be able to navigate away from an in-progress job and return to see its current state.

4.4 Performance & Scalability (documented constraints, not guaranteed SLAs in V1)

- NFR-4.1: Job state is held in-process (in-memory), not in a persistent queue; this is an accepted V1 limitation (see §7).
- NFR-4.2: Login rate-limiting is tracked per process; multi-process deployment proportionally raises the effective attempt ceiling (accepted V1 limitation).

4.5 Compliance / Localization

- NFR-5.1: All monetary values shall be presented in INR with GST-compliant invoice fields (GSTIN, tax breakdown).

5. External Interfaces

5.1 API

A REST API under `/api`, consumed by the bundled React frontend and by internal QA tooling (Postman/JMeter) via the `X-API-Key` mechanism. Endpoint groups: Auth, Jobs, Estimations/Documents, Invoices, Organization, Rate Card, Analytics, Config/Health.

5.2 Third-Party AI Services

- **Mistral** — OCR of PDF requirement documents.
- **Groq** — LLM analysis, estimation, and BRD/SRS text generation.
- **Google Gemini** — web-grounded market pricing research.

The system shall support multiple API keys per provider with automatic rotation/failover on rate-limit errors.

5.3 User Interface

A single-page React application with two primary workspaces reachable from a landing page: **Estimation Maker** and **Invoice Maker**, each with its own dashboard, creation flow, history list, detail view, and shared organization settings page.

6. Data Requirements

Core persisted entities: `User`, `OrganizationProfile`, `RateCard`, `Client`, `Estimation`, `Document` (versioned Quotation/BRD/SRS content), `Invoice`, `AuditLog`, `BrandingAsset`. `Estimation`, `Invoice`, and `Document` numbers are generated as year-scoped sequential identifiers (e.g. `EST-2026-000123`, `INV-2026-000045`).

7. Known V1 Limitations / Explicit Non-Goals

These are documented so they are not mistaken for defects during QA, and so they can be scoped deliberately for V2:

1. **No role-based access control (RBAC).** Any authenticated user has full access to every action in the system, including organization/branding, rate card, and estimation/invoice edits and deletes. This is a candidate for future development — e.g. an Admin/Finance/PM/Developer role split — if the business later needs separation of duties; it is not required for V1.
2. **No horizontal scaling of the AI pipeline.** Job state lives in an in-process dictionary; it does not survive a server restart and is not shared across multiple worker processes.
3. **No account deactivation flag.** A user account can be deleted but not temporarily disabled.
4. **No server-side token revocation.** A leaked session token remains valid until its 24-hour natural expiry.
5. **No self-service client portal.** All access is internal-staff-only; clients receive documents as exported PDFs, not through an account.

6. **No payment gateway integration.** Invoice "Paid" status is recorded manually, not verified against a payment processor.
7. **Single currency (INR) and single tax regime (Indian GST).** No multi-currency or multi-region support.
8. **No formal schema migration framework.** Schema changes are additive (`ALTER TABLE ADD COLUMN`) rather than managed by a tool such as Alembic; this is flagged in the codebase as a pre-scale-up gap.
9. **Attachment data model exists but has no wired endpoints** — treated as reserved for a future release, not a V1 feature.
10. **Two diagnostic/debug endpoints remain in the API surface** (an organization-profile debug dump and a PDF smoke-test route) and should be removed or explicitly access-controlled before external exposure.

8. Glossary

Term	Meaning
BRD	Business Requirements Document
SRS	Software Requirements Specification (the generated client document, distinct from this file)
GST	Goods and Services Tax (India)
GSTIN	GST Identification Number
OCR	Optical Character Recognition
SSRF	Server-Side Request Forgery
Optimistic locking	Concurrency control via a version counter, rejecting stale writes instead of blocking readers